TECHNOLOGY PORTFOLIO CV

Gene Boo Ee Jin

Quantitative Finance, Mathematics & Computational Systems Architect

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Python | WebGL | Three.js | LLMs | Fourier | Parquet / SQL | Rust / C++ | DSP

PROFILE

Quantitative finance practitioner and computational systems builder with 25+ years across derivatives, risk analytics, numerical methods, model validation, balance sheet risk, software engineering and creative computing.

Personal and research projects extend into WebGL, Three.js, Python, Rust, C++, REST services, audio/DSP, generative visuals, educational math tools and browser-based computational interfaces.

25+
Years across risk, quant and systems

3
Core domains: finance, math, software

A4
Readable on screen and print

PROFESSIONAL EXPERIENCE

IBMR Projects Consultant
AmBank Group · Mar 2025 – Present

- Built governed IRRBB behavioural modelling engines for CASA/NMD and FD/TDRR portfolios under BNM-aligned methodology.
- Implemented out-of-core Python/SQL pipelines using Polars, DuckDB, pandas, NumPy, Parquet, YAML and SQL Server/T-SQL.
- Developed CASA/NMD logic for stable balances, core/non-core decomposition, PTR calibration, BNM caps, WAM schedules, 19-bucket tenor grids, shocks and EVE attribution.
- Built scalar PTR and PCA/PCR-vector PTR challenger approaches with curve governance, missing-rate treatment, audit visibility and reporting.
- Built FD/TDRR early-redemption models using monthly account-history reconstruction, cohort calibration, scenario-scaled redemption ratios and contractual maturity scheduling.

Product Specialist
Qontigo / AxiomaRisk / SimCorp · Dec 2021 – Jun 2023

- Delivered quantitative product support for AxiomaRisk and performance attribution across buy-side multi-asset workflows.
- Built Python/API automation for position loading, report extraction, validation and Excel analytics.
- Explained factor models, VaR, ES, liquidity analytics, PCA modes and multi-asset risk outputs to clients.

Senior Quantitative Analyst
Maybank Group · Sep 2019 – Aug 2021

- Developed Monte Carlo, bootstrap, copula and Euler allocation models for operational-risk capital.
- Built FFT convolution and distribution-fitting tools in Python for arbitrary empirical distributions.
- Created COVID-19 R0 estimation models and automated enterprise risk templates.

Head of Market Risk Model Validation
Maybank Group · Mar 2014 – Mar 2019

- Validated derivative pricing, valuation and risk models across FX, IR, equity, commodity, credit, liquidity and IRRBB domains.
- Developed independent pricing and validation engines using Python, C++/XLL, VBA, R and SQL-connected sources.
- Authored documentation standards, workflow policies, validation thresholds and governance frameworks.

Senior Quantitative Analyst
AmBank Group · Apr 2009 – Mar 2014

- Developed benchmark pricing models, volatility-surface builders, correlation tools and derivative analytics for FX, equity and interest-rate products.
- Implemented practical quant utilities in C++, Python and VBA for valuation review, sensitivities, market-data treatment and model behaviour analysis.
- Supported market-risk methodology and quant coaching, translating derivative concepts into maintainable tools for day-to-day risk and validation workflows.

Risk Specialist
Pilot Multimedia · Jun 2007 – Jan 2008

- Built XML staging tools and VB/VBA plug-ins for a proprietary C# credit-risk application, connecting formula definitions to executable model components.
- Implemented logistic-regression based probability-of-default utilities and supported translation of credit-risk concepts into software workflows.
- Worked across risk logic, data staging and application design, strengthening the bridge between quantitative methodology and production-style tooling.

Market Risk Analyst
Fortis Bank / BNP Paribas Fortis · Aug 2006 – Jan 2007

- Developed Asian Basket option pricing and Greeks models in C++ and Python, combining analytical approximation, FFT and Monte Carlo methods.

- Implemented and compared Deelstra, Curran, Ju, Gram-Charlier and simulation-based techniques to understand pricing behaviour and sensitivity profiles.
- Built quantitative prototypes for exotic-option market-risk review, with emphasis on numerical approach, performance and transparent model behaviour.

Contracts Analyst

Electrabel / ENGIE Electrabel · Aug 2000 – May 2006

- Developed spark-spread and crack-spread pricers in VBA/C++ for energy derivative workflows, connecting contract economics to valuation and risk treatment.
- Supported energy derivatives integration into Murex-related risk processes, helping align contract representation, pricing tools and risk-engine treatment.
- Built SAS-based hourly power-profile tools using Winter-method approaches, supporting load-shape analysis and power-market exposure modelling.

EDUCATION & CERTIFICATIONS

MBA, Global Management

Hochschule Bremen, Germany · Sep 1999 · Grade 1.0

BA, Business Administration

University of Hertfordshire, UK · Mar 1998 · 2nd Upper Honours

AICB Risk Management

Principles & Framework · Risk Models, Capital & ALM

Microsoft Visual Basic 6

Certification

SIGNATURE TOOLKIT

Quant Finance

Derivative pricing · market risk · operational risk · IRRBB · EVE · SA-CCR · VaR / ES · Euler allocation · model validation · ALM

Computational Math

Monte Carlo · copulae · FFT · convolution · PCA / PCR · distribution fitting · KDE · quadrature · local volatility · optimisation

Software & Data

Python · NumPy · SciPy · pandas · Polars · DuckDB · mpmath · SQL Server · MySQL · Parquet · YAML · C++ · Rust · REST APIs

Creative Computing

JavaScript · HTML · CSS · WebGL · Three.js · SVG · Canvas · Fourier epicycles · audio-reactive systems · DSP experiments

AI Build Suite

Copilot · ChatGPT · Gemini · Grok · DeepSeek · Kimi · Ollama · Hugging Face · local LLM inference · Cursor · AI-assisted analytics

RESEARCH NOTES

Understanding Fourier Transform
Fourier foundations, convolution-based option pricing and Python implementation.

FFT Cross Correlation vs Pearson Correlation
FFT-based correlation methods compared with Pearson correlation.

Analytical Black-Scholes-like Pricing
Moment-generating-function drift correction under risk-neutral valuation.

Mind the Gap (Under the Curve)
Numerical integration, quadrature and financial engineering applications.

Art of Buy-Side Arbitrage Strategy
Quantitative perspective on buy-side arbitrage operations.

CUDA / Matrix Multiplication Notes
CUDA tiling and high-performance numerical computing explorations.

SELECTED PROJECTS

ExcelBridge
Zero-install spreadsheet analytics connected to server-side Python numerical computing.

Fit_it!
Distribution fitting and KDE platform for empirical data analysis.

FunkyGraffy
Interactive mathematical visualization and edutainment platform.

FourierDoodler
Browser-based Fourier epicycle drawing and animation tool.

Convolve!
Empirical convolution tool for arbitrary distributions and joint-distribution modelling.

Gene's SaaS Universe
Mini REST endpoints for derivative pricing, quant analytics, Kalman filtering and forecasting.

Large Parquet Viewer / Rust-Polars tools
Lightweight tooling for exploring large parquet datasets with SQL/PRQL-style queries.

WebGL Audio Visualizer
Real-time Three.js/Web Audio visual system for GPU-accelerated reactive visuals.

Sonic Art / DSP Experiments
C++/Rust/JUCE learning projects around synthesizers, mastering plugins and generative audio-visual systems.

Private Security Concepts
PersVa encrypted storage, AES/XOR file obfuscation and VPN encryption experiments.

LANGUAGES

English: Native · Mandarin: Fluent · German: Intermediate · French: Basic · Flemish: Basic · Cantonese: Basic · Malay: Basic.